

NILPOTENCY THRESHOLD AND LOEWY LENGTH OF THE KALEIDOSCOPE YANG–BAXTER ALGEBRA

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ABSTRACT. The consistency condition of the coordinate Bethe ansatz on a fan of δ -function mirrors is a single matrix identity, and it generates a finite-dimensional algebra spanned by the words in two matrices: a clock matrix T of order N and a matrix Λ of square zero built from the quantum torus at the N th root of unity, both depending on a spectral parameter u . Whether every word in which at least $N/2$ powers of T separate consecutive copies of Λ vanishes was left open, with verification reported only through $N = 10$. We prove it for every even $N = 2n \geq 4$, every u with $u^N \neq 1$ and every tuple of integer exponents. The mechanism is a flag of subspaces obtained by interpolating separately on the even and the odd values of $u\omega^j + u^{-1}\omega^{-j}$: the operator $t + t^{-1}$ preserves this flag, every operator $s(t^k - t^{-k})$ lowers it by one step, and a Chebyshev factorization makes the second statement uniform in the integer k . The threshold is sharp. The word with $n - 1$ separating factors, all equal to T , has rank two at every admissible u , has poles exactly at $u^N = 1$ and vanishes nowhere else. Consequently the two-sided ideal J generated by Λ satisfies $J^{n+1} = 0$ and $J^n \neq 0$ and is the Jacobson radical, so the algebra has $n + 1$ nonzero Loewy layers and Loewy length $N/2 + 1$, one more than the value proposed with the conjecture, which is instead the largest exponent with $J^n \neq 0$. Five properties of the data carry the whole argument; taken as axioms over an arbitrary field they give the same threshold, the same sharpness and the same radical, and for each axiom there is an explicit datum over $\mathbb{Q}$ that satisfies the other four and violates a conclusion the deleted axiom supports.

1. INTRODUCTION

A quantum particle in the plane scattering off a fan of δ -function mirrors through the origin is one of the few two-dimensional problems that a coordinate Bethe ansatz solves exactly. When the mirror lines carry the dihedral group of a regular N -gon they cut the plane into $2N$ wedges, and inside each wedge the eigenfunction is a finite superposition of plane waves; the mirrors relate the amplitudes of neighbouring wedges. These are Gaudin’s generalized kaleidoscopes [1, Ch. 5], named for the mirror arrangements of classical geometry [2], and the reflection groups that classify them are the finite reflection groups [3]. The construction descends from the one-dimensional Bethe ansatz [4–10], and it has been revived by the asymmetric Bethe ansatz of Jackson, Perrin, Astrakharchik and Olshanii [11, 12], which restricts a kaleidoscope solution to the sector odd under the generating mirrors of a reflection subgroup [13] and thereby solves models, such as the mass-imbalanced pair in a hard-wall trap [14], that appear to violate the classical integrability conditions.

Qiu, Guan and Yu [15] showed that the consistency of the ansatz on the whole fan — the requirement that an amplitude transported once around the full circuit of mirrors return to itself — is one matrix identity, which they call the kaleidoscope Yang–Baxter equation. Fix an even integer $N = 2n \geq 4$ and let $\omega = e^{2\pi i/N}$. On the standard basis $e_0, \dots, e_{N-1}$ of $\mathbb{C}^N$, let t be the cyclic shift $te_j = e_{j-1 \bmod N}$ and let d be the clock matrix $de_j = \omega^j e_j$, so that $t^N = d^N = I$ and $td = \omega dt$: the two generate the quantum torus at the root of unity ω , an algebra that appears as a subalgebra in the theory of quantum groups [16] and that carries the magnetic translations of the

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quantum Hall effect [17] and the Fourier modes of noncommutative field theory [18]. The spectral parameter is a complex number $u = e^{i\theta}$ recording the incidence angle, and the two matrices that carry the scattering are

$$T = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}, \quad \Lambda = \begin{pmatrix} s & s \\ -s & -s \end{pmatrix}, \quad s = (ud - u^{-1}d^{-1})^{-1}, \quad (1.1)$$

acting on $\mathbb{C}^{2N}$; here T transports an amplitude past one mirror and Λ carries the strength of the δ -interaction, and the diagonal matrix s is defined wherever $u^N \neq 1$. The kaleidoscope Yang–Baxter equation is the statement that $[T(I + z\Lambda)]^N = I$ for every z , and expanding it in z yields $T^N = I$, $\Lambda^2 = 0$ and $(T\Lambda)^N = 0$.

A fourth statement was left as a conjecture [15]: that every word alternating powers of T with copies of Λ dies once enough powers of T intervene,

$$\Lambda T^{k_1} \Lambda T^{k_2} \cdots \Lambda T^{k_M} \Lambda = 0 \quad \text{whenever } M \geq \frac{N}{2}, \quad (1.2)$$

for arbitrary integers $k_1, \dots, k_M$. Its authors report that they could not prove it and verified it only for $N \leq 10$. The obstacle is not the identity $\Lambda^2 = 0$, which handles two adjacent copies of Λ and nothing more: each intervening power of T moves the image of the next Λ , the exponents are unrestricted integers, and no finite check reaches every N . On the strength of the conjecture the same work describes the algebra generated by T and Λ as splitting into a semisimple part spanned by the powers of T and a nilpotent ideal generated by Λ , and assigns it Loewy length $N/2$.

We prove Eq. (1.2), prove that its threshold cannot be lowered, and settle the structural statement that rests on it. Writing $\mathcal{U}_N = \{u \in \mathbb{C}^\times : u^N \neq 1\}$ for the set of spectral parameters at which s is defined, and $\mathcal{A}$ for the unital algebra generated by T and Λ inside $M_{2N}(\mathbb{C})$, the three statements are the following.

Theorem 1.1 (Threshold and sharpness). *Let $N = 2n \geq 4$ be even and let $u \in \mathcal{U}_N$. Then*

$$\Lambda T^{k_1} \Lambda \cdots \Lambda T^{k_M} \Lambda = 0 \quad \text{for every } M \geq n \text{ and every } (k_1, \dots, k_M) \in \mathbb{Z}^M.$$

At $M = n - 1$ the word with $k_1 = \cdots = k_{n-1} = 1$ has rank two for every $u \in \mathcal{U}_N$. As a matrix of rational functions of u it has poles exactly at the N th roots of unity, and there is no $u \in \mathcal{U}_N$ at which it vanishes.

Theorem 1.2 (Radical and Loewy length). *Let $J = \mathcal{A}\Lambda\mathcal{A}$. Then $J = \text{rad } \mathcal{A}$, $J^{n+1} = 0$ and $J^n \neq 0$, every inclusion in $\mathcal{A} \supsetneq J \supsetneq \cdots \supsetneq J^n \supsetneq J^{n+1} = 0$ is strict, and the Loewy length of $\mathcal{A}$ is $n + 1 = N/2 + 1$.*

The value $N/2$ proposed with the conjecture is therefore the largest exponent r with $J^r \neq 0$, and not the number of Loewy layers; the two differ by one, and Theorem 1.2 fixes which is which.

The kaleidoscope condition sits beside, rather than inside, the two familiar consistency relations of integrable scattering. The Yang–Baxter equation equates the two orders in which three components may scatter pairwise, and the reflection equation of Cherednik [19] and Sklyanin [20] adds a boundary; the kaleidoscope condition instead closes a circuit of $2N$ wedges, and the algebra it generates is nilpotent above its semisimple part rather than semisimple. A different route to delta-interactions with reflection symmetry runs through degenerate double affine Hecke algebras and Dunkl-type operators [21]; it yields the Bethe ansatz equations and eigenfunctions rather than a finite matrix algebra, and does not reach the question settled here.

The proof of Theorem 1.1 occupies Sections 2 to 5 and uses one mechanism. A similarity that is independent of every exponent turns the $2N \times 2N$ word into a single $N \times N$ word in the quantum torus, built from the operators $A_k = s(t^k - t^{-k})$. On $\mathbb{C}^N$ we construct a flag $0 = K_0 \subset K_1 \subset \cdots \subset K_n = \mathbb{C}^N$ whose r th member consists of the vectors interpolated, separately on the even and the odd indices j , by polynomials of degree less than r in $y_j = u\omega^j + u^{-1}\omega^{-j}$. The operator $t + t^{-1}$ preserves each K_r and A_1 maps K_r into K_{r-1} ; and because $t^k - t^{-k} = (t - t^{-1})q_k(t + t^{-1})$ for an

integer Chebyshev polynomial q_k , so does A_k , for every integer k at once. A word with n such factors therefore carries $\mathbb{C}^N = K_n$ into $K_0 = 0$, and the degree lowering is exact enough that one factor fewer leaves a rank-two matrix.

The mechanism uses five properties of the pair (t, s) and the flag, and nothing else. [Section 7](#) takes those five as axioms over an arbitrary field, proves the threshold, the sharpness and the radical statement from them, and shows that each axiom is indispensable by exhibiting, for each one, an explicit datum over $\mathbb{Q}$ that satisfies the other four and breaks a conclusion that axiom supports. [Section 8](#) reports exact computations run against the original definitions as a check on the conventions.

2. THE ALGEBRA AND ITS SCALAR REDUCTION

Throughout, $N = 2n \geq 4$ is even, $\omega = e^{2\pi i/N}$ is a primitive N th root of unity, $V = \mathbb{C}^N$ carries the shift and clock matrices

$$te_j = e_{j-1 \bmod N}, \quad de_j = \omega^j e_j \quad (j \in \mathbb{Z}/N\mathbb{Z}), \quad (2.1)$$

and T , Λ and s are as in [Eq. \(1.1\)](#). Reading [Eq. \(2.1\)](#) on components, t moves the entries of a vector one place down the cycle and d multiplies the j th entry by ω^j ; the two therefore commute up to the fixed phase ω .

The matrix s is diagonal with entries

$$s_j = (u\omega^j - u^{-1}\omega^{-j})^{-1}, \quad (2.2)$$

so s fails to exist exactly where some entry of $ud - u^{-1}d^{-1}$ vanishes.

Lemma 2.1 (Admissible domain). *The matrix s is defined and invertible precisely for $u \in \mathcal{U}_N = \{u \in \mathbb{C}^\times : u^N \neq 1\}$.*

Proof. The j th entry of $ud - u^{-1}d^{-1}$ vanishes if and only if $u^2\omega^{2j} = 1$. Raising that equation to the power n and using $\omega^N = 1$ gives $u^N = 1$. Conversely, if $u^N = 1$ then u^2 is an n th root of unity, hence equals ω^{-2j} for some j because ω^2 has order n . Where every entry is nonzero, s is a diagonal matrix with nonzero entries. $\square$

Every statement below is made for $u \in \mathcal{U}_N$, and $T^N = I$ lets any integer exponent on T be reduced modulo N without changing a word.

The block structure of T and Λ can be removed once and for all.

Lemma 2.2 (Scalar reduction). *Put*

$$S = \begin{pmatrix} I & I \\ -I & 0 \end{pmatrix}, \quad S^{-1} = \begin{pmatrix} 0 & -I \\ I & I \end{pmatrix}, \quad A_k = s(t^k - t^{-k}). \quad (2.3)$$

Then for every $k \in \mathbb{Z}$

$$S^{-1}T^k S = \begin{pmatrix} t^{-k} & 0 \\ t^k - t^{-k} & t^k \end{pmatrix}, \quad S^{-1}\Lambda S = \begin{pmatrix} 0 & s \\ 0 & 0 \end{pmatrix}, \quad (2.4)$$

and for all $M \geq 1$ and $(k_1, \dots, k_M) \in \mathbb{Z}^M$

$$S^{-1}(\Lambda T^{k_1} \Lambda \dots \Lambda T^{k_M} \Lambda) S = \begin{pmatrix} 0 & A_{k_1} A_{k_2} \dots A_{k_M} s \\ 0 & 0 \end{pmatrix}. \quad (2.5)$$

Proof. Both identities in [Eq. \(2.4\)](#) are a single block multiplication. In the transformed basis Λ has only the upper-right block s , so a product of the two transformed matrices keeps only the upper-right block, and each factor $S^{-1}T^{k_i} S$ contributes its lower-left block $t^{k_i} - t^{-k_i}$ sandwiched between the two neighbouring copies of s . Iterating gives [Eq. \(2.5\)](#). $\square$

Equation (2.5) says that the $2N \times 2N$ conjecture is exactly an $N \times N$ problem: the chained word vanishes if and only if the single product $A_{k_1} \cdots A_{k_M} s$ vanishes on V . Only the lower-left block of the transformed T^k ever enters, so the diagonal blocks play no part in what follows.

Remark 2.3. The reference that poses the conjecture prints the lower-right block of the transformed T^p as t^{-p} ; direct conjugation gives t^p , as in Eq. (2.4). Nothing depends on the difference, because only the lower-left block enters Eq. (2.5).

3. THE PARITY-POLYNOMIAL FLAG

Fix $u \in \mathcal{U}_N$ and set

$$z_j = u\omega^j, \quad y_j = z_j + z_j^{-1} \quad (j \in \mathbb{Z}/N\mathbb{Z}).$$

The numbers y_j are the interpolation nodes of everything below, and the parity of j separates them into two classes of n nodes each, because the shift t exchanges even and odd indices while two shifts return to the class they started in. In this notation the diagonal entries of s from Eq. (2.2) are $s_j = (z_j - z_j^{-1})^{-1}$.

Lemma 3.1 (Distinct nodes within each parity). *For $u \in \mathcal{U}_N$ the n numbers y_j with j even are pairwise distinct, and so are the n numbers y_j with j odd.*

Proof. For nonzero a, b the equality $a + a^{-1} = b + b^{-1}$ forces $a = b$ or $ab = 1$. Apply this to $a = z_j$ and $b = z_\ell$ for distinct indices j, ℓ of the same parity. The case $a = b$ gives $\omega^{j-\ell} = 1$ with $0 < |j - \ell| < N$, contradicting the primitivity of ω . The case $ab = 1$ gives $u^2\omega^{j+\ell} = 1$; raising it to the power n and using $\omega^n = -1$ yields $u^N = (-1)^{j+\ell} = 1$, since $j + \ell$ is even, which contradicts $u \in \mathcal{U}_N$. $\square$

Nodes of opposite parity may coincide, and nothing below excludes it: the two classes are interpolated independently.

Definition 3.2 (The flag). For $0 \leq r \leq n$ let $K_r \subseteq V$ be the set of vectors f for which there exist polynomials P_0, P_1 with $\deg P_0 < r$ and $\deg P_1 < r$ such that

$$f_j = \begin{cases} P_0(y_j), & j \text{ even}, \\ P_1(y_j), & j \text{ odd}, \end{cases}$$

with $K_0 = 0$.

A vector of K_r is thus described by two independent polynomials, one for each parity class, of degree less than r in the node variable.

Lemma 3.3 (Dimensions). $0 = K_0 \subset K_1 \subset \cdots \subset K_n = V$ and $\dim K_r = 2r$.

Proof. By Lemma 3.1 the n nodes of a fixed parity are distinct, so on that class the map sending a polynomial of degree less than $r \leq n$ to its values is injective, and for $r = n$ it is a bijection onto all functions on the class. Summing the two classes gives $\dim K_r = 2r$ and $K_n = V$. $\square$

Two operators act on this flag:

$$A = s(t - t^{-1}), \quad X = t + t^{-1}. \quad (3.1)$$

The first is the $k = 1$ member of the family A_k of Lemma 2.2 and carries the interaction; the second is the symmetric combination of the two shifts and carries no s . On components, using $(tf)_j = f_{j+1}$ and $(t^{-1}f)_j = f_{j-1}$,

$$(Af)_j = \frac{f_{j+1} - f_{j-1}}{z_j - z_j^{-1}}, \quad (Xf)_j = f_{j+1} + f_{j-1}. \quad (3.2)$$

Both operators read the two neighbours of the index j , which lie in the parity class opposite to j ; A takes their difference and divides by the same quantity that defines s_j , while X takes their sum.

Lemma 3.4 (Propagation and lowering). *For $1 \leq r \leq n$,*

$$XK_r \subseteq K_r, \quad AK_r \subseteq K_{r-1}.$$

Moreover, if the polynomial representing f on one parity class has degree m and leading coefficient c , then the polynomial representing Af on the opposite class has degree $m-1$ and leading coefficient

$$c(\omega^m - \omega^{-m}), \quad (3.3)$$

which is nonzero for $1 \leq m \leq n-1$.

Proof. Fix j , write $z = z_j$, and set $a = y_{j+1}$ and $b = y_{j-1}$. Expanding,

$$a - b = (\omega - \omega^{-1})(z - z^{-1}), \quad a + b = (\omega + \omega^{-1})(z + z^{-1}), \quad ab = (z + z^{-1})^2 + (\omega - \omega^{-1})^2. \quad (3.4)$$

The two elementary symmetric functions of a and b are therefore polynomials in $y_j = z + z^{-1}$ of degrees one and two respectively, while their difference carries the factor $z - z^{-1}$ that Eq. (3.2) divides by.

Let P be the polynomial representing f on the parity class of $j \pm 1$, so $f_{j+1} = P(a)$ and $f_{j-1} = P(b)$. The sum $P(a) + P(b)$ is a symmetric polynomial in a, b , hence a polynomial in $a + b$ and ab of weighted degree at most $\deg P$; by Eq. (3.4) it is a polynomial in y_j of degree at most $\deg P$. This proves $XK_r \subseteq K_r$. Likewise the divided difference $(P(a) - P(b))/(a - b)$ is symmetric of weighted degree at most $\deg P - 1$, hence a polynomial in y_j of degree at most $\deg P - 1$; since

$$(Af)_j = \frac{P(a) - P(b)}{z - z^{-1}} = (\omega - \omega^{-1}) \frac{P(a) - P(b)}{a - b}, \quad (3.5)$$

the operator A drops the degree by at least one, which proves $AK_r \subseteq K_{r-1}$. The middle expression in Eq. (3.5) is what A computes, and the right-hand one exhibits it as a divided difference in the node variable times a constant.

For the leading coefficient take $P(x) = cx^m$ plus lower-order terms. Then $(P(a) - P(b))/(a - b) = c \sum_{i=0}^{m-1} a^i b^{m-1-i}$ plus terms of lower degree. Letting $z \rightarrow \infty$ in $a \sim \omega z$, $b \sim \omega^{-1}z$ and $y_j \sim z$, the sum behaves as $z^{m-1} \omega^{-(m-1)} (\omega^{2m} - 1)/(\omega^2 - 1)$, so together with the prefactor in Eq. (3.5) the coefficient of y_j^{m-1} is $c(\omega^m - \omega^{-m})$. It vanishes only when $\omega^{2m} = 1$, that is when n divides m , which is excluded for $1 \leq m \leq n-1$. $\square$

The degree drop is thus not merely an inequality: below the ceiling n it is exact, and the constant by which the leading coefficient is multiplied never vanishes. That is what forces the sharpness in Section 5, and it also pins down the Jordan type of A .

Corollary 3.5 (Jordan type of A). *For $0 \leq r \leq n$,*

$$\ker A^r = K_r, \quad \text{rank } A^r = N - 2r. \quad (3.6)$$

In particular $A^n = 0$ and A has exactly two nilpotent Jordan blocks, each of size n .

Proof. By Lemma 3.3 every $f \in V$ is represented by a unique pair of polynomials of degree less than n , and Lemma 3.4 gives $K_r \subseteq \ker A^r$. Conversely suppose one of the two polynomials has degree m with $r \leq m \leq n-1$. Applying A once transfers it to the opposite class with degree $m-1$ and a nonzero leading coefficient, by Eq. (3.3); the two classes never mix, so after r steps a polynomial of degree $m-r \geq 0$ with nonzero leading coefficient survives and $A^r f \neq 0$. Hence $\ker A^r = K_r$, and Lemma 3.3 gives the rank. The kernel dimensions increase by two at each step up to $r = n$, which is the Jordan type stated. $\square$

4. UNIFORM VANISHING FOR ARBITRARY EXPONENTS

Lemma 3.4 lowers the flag by the single operator $A = A_1$. The step that makes the argument uniform in the exponents is that every A_k factors through A .

Lemma 4.1 (Integer Chebyshev factorization). *Define $q_k \in \mathbb{Z}[x]$ by*

$$q_0 = 0, \quad q_1 = 1, \quad q_{k+1} = x q_k - q_{k-1} \quad (k \geq 1), \quad q_{-k} = -q_k. \quad (4.1)$$

Then for every $k \in \mathbb{Z}$

$$t^k - t^{-k} = (t - t^{-1}) q_k(X), \quad \text{hence} \quad A_k = A q_k(X). \quad (4.2)$$

Proof. Set $Y_k = t^k - t^{-k}$. Since t commutes with t^{-1} and hence with X ,

$$XY_k = (t + t^{-1})(t^k - t^{-k}) = Y_{k+1} + Y_{k-1},$$

so $Y_{k+1} = XY_k - Y_{k-1}$, which is the recurrence in [Eq. \(4.1\)](#). The cases $k = 0$ and $k = 1$ of [Eq. \(4.2\)](#) are immediate, and induction gives every $k \geq 0$; replacing k by $-k$ gives the negative ones, since both sides change sign. Multiplying on the left by s turns the first identity into the second. $\square$

The polynomials q_k are the Chebyshev polynomials of the second kind in a normalization that avoids division: $q_k(x) = U_{k-1}(x/2)$ for $k \geq 1$ [\[22\]](#). The recurrence in [Eq. \(4.1\)](#) has integer coefficients, so [Lemma 4.1](#) holds over any commutative ring and in particular in every characteristic.

Proposition 4.2 (Uniform lowering). *For every $k \in \mathbb{Z}$ and every $1 \leq r \leq n$,*

$$A_k K_r \subseteq K_{r-1}. \quad (4.3)$$

Proof. By [Lemma 3.4](#) the operator X preserves every K_r , hence so does any polynomial in X ; and A lowers the flag by one. Now apply [Eq. \(4.2\)](#). $\square$

[Equation \(4.3\)](#) is the whole content of the vanishing half of [Theorem 1.1](#): no matter which integer the exponent is — positive, zero, negative, or outside any chosen residue system — the corresponding operator moves a vector exactly one step down the same flag.

Proof of the vanishing statement of Theorem 1.1. By [Lemma 2.2](#) the word $\Lambda T^{k_1} \Lambda \cdots \Lambda T^{k_M} \Lambda$ vanishes if and only if $A_{k_1} \cdots A_{k_M} s$ does. Read that product from the right, as in [Fig. 1](#). Since s is invertible on $\mathcal{U}_N$ by [Lemma 2.1](#), $sV = V = K_n$ by [Lemma 3.3](#). Each of the M factors A_{k_i} then lowers the flag by one step, by [Proposition 4.2](#). If $M \geq n$ the image lies in $K_0 = 0$. $\square$

No ordering, positivity, genericity or congruence condition on the exponent tuple was used, and N entered only through the length n of the flag.

5. SHARPNESS AND THE POLE LOCUS

One factor fewer does not suffice, and the failure is as large as it can be.

Proposition 5.1 (The sharp word). *Let $u \in \mathcal{U}_N$ and take $M = n - 1$ with $k_1 = \cdots = k_{n-1} = 1$. Then the scalar block of the corresponding word is $A^{n-1}s$ and*

$$\text{rank}(A^{n-1}s) = 2.$$

In particular the word is nonzero at every admissible spectral parameter, not merely at generic ones.

Proof. Define $f \in V$ by $f_j = y_j^{n-1}$ for j even and $f_j = 0$ for j odd. Applying A repeatedly and using the exact degree drop [Eq. \(3.3\)](#) at $m = n - 1, n - 2, \dots, 1$,

$$A^{n-1}f = \left[\prod_{m=1}^{n-1} (\omega^m - \omega^{-m}) \right] \mathbf{1}_\epsilon, \quad (5.1)$$

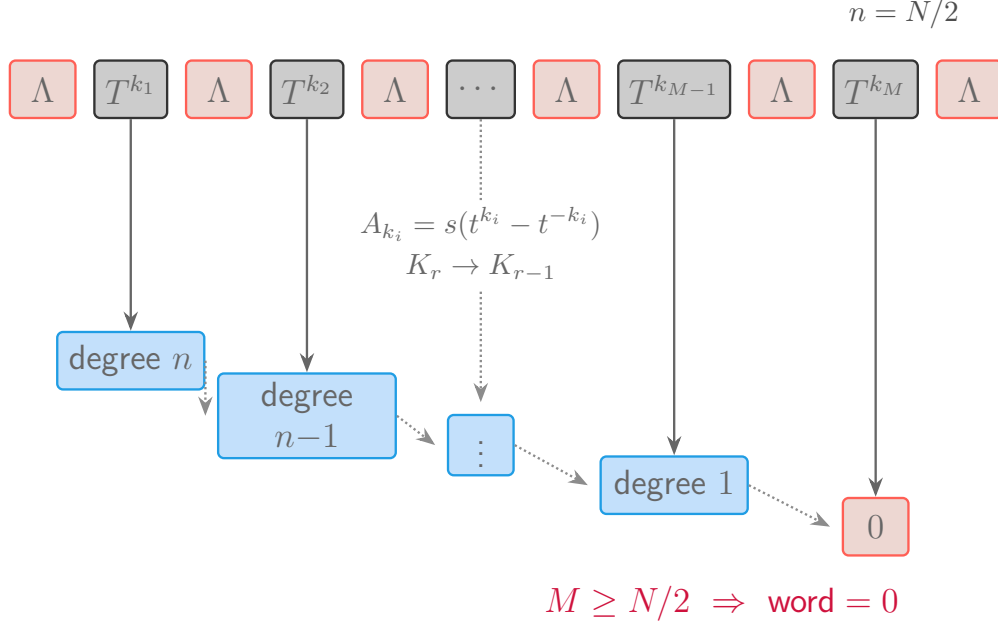


FIGURE 1. How a chained word is forced to zero. Reading the word from the right, the invertible factor s maps $\mathbb{C}^N$ onto the top member K_n of the flag, and each intervening power T^{k_i} contributes one operator $A_{k_i} = s(t^{k_i} - t^{-k_i})$ that carries K_r into K_{r-1} , independently of the integer k_i . After $n = N/2$ such steps the image lies in $K_0 = 0$.

where $\mathbf{1}_\epsilon$ is the indicator vector of the parity class reached after $n - 1$ transfers. Every factor of the product is nonzero, so $A^{n-1}f \neq 0$; applying $A^{n-1}s$ to $s^{-1}f$ shows $A^{n-1}s \neq 0$. The rank follows from Eq. (3.6) at $r = n - 1$ together with the invertibility of s . $\square$

Equation (5.1) says that the vector supported on one parity class with values the top power of the node variable is pushed, in $n - 1$ exact steps, to a nonzero constant vector on the other class.

It remains to locate the singularities of the sharp word as a function of the spectral parameter, and to check that the points excluded from $\mathcal{U}_N$ are genuine poles rather than removable ones.

Proposition 5.2 (Poles and zeros of the sharp word). *As a matrix of rational functions of u , the sharp word $A^{n-1}s$ has poles exactly at the N th roots of unity, each simple in at least one entry, and it has no zero in $\mathcal{U}_N$.*

Proof. Absence of zeros in $\mathcal{U}_N$ is Proposition 5.1. For the poles, read $A = s(t - t^{-1})$ in matrix entries: its only nonzero entries are $A_{j-1,j} = s_{j-1}$ and $A_{j+1,j} = -s_{j+1}$. Fix a column ℓ and consider the entry of A^{n-1} in row $\ell - (n - 1)$. A contributing path takes $n - 1$ steps of ± 1 and must have total displacement $-(n - 1)$ modulo N ; the only alternative displacement, $-(n - 1) + N = n + 1$, exceeds the number of steps, so all steps go the same way and the path is unique. Multiplying by the diagonal s on the right,

$$(A^{n-1}s)_{\ell-(n-1),\ell} = \prod_{r=0}^{n-1} s_{\ell-r}. \quad (5.2)$$

This entry is a product of n consecutive diagonal entries of s , with no competing path to cancel it.

Let $u_0^N = 1$. By the proof of Lemma 2.1 the entries s_j that blow up at u_0 are those with $u_0^2 \omega^{2j} = 1$, and there are exactly two of them, at indices differing by n . Choose ℓ to be one of the two. The n consecutive indices $\ell, \ell - 1, \dots, \ell - (n - 1)$ contain that index and not the other, which is n steps away. The corresponding denominator $u\omega^\ell - u^{-1}\omega^{-\ell}$ has derivative $2\omega^\ell \neq 0$ at

u_0 , hence a simple zero there, while all the remaining factors in Eq. (5.2) are finite and nonzero. The entry therefore has an uncanceled simple pole at u_0 . Since every s_j is finite on $\mathcal{U}_N$, there is no pole elsewhere. $\square$

Propositions 5.1 and 5.2 complete Theorem 1.1: the threshold $M \geq n$ cannot be replaced by $M \geq n - 1$, the counterexample family is explicit, and its only excluded parameters are exactly the ones at which the construction is undefined to begin with.

6. THE RADICAL AND THE LOEWY LENGTH

Let $\mathcal{A} \subseteq M_{2N}(\mathbb{C})$ be the unital algebra generated by T and Λ , and let

$$J = \mathcal{A}\Lambda\mathcal{A}$$

be the two-sided ideal generated by Λ , that is, the span of all words in $T^{\pm 1}$ and Λ containing at least one Λ .

Proof of Theorem 1.2. Expanding a product of $n + 1$ elements of J gives a linear combination of words each containing at least $n + 1$ copies of Λ . In such a word the letters between two consecutive copies of Λ are powers of T , which combine into a single T^k with $k \in \mathbb{Z}$, an empty gap giving $k = 0$. Any $n + 1$ consecutive copies of Λ therefore form a subword $\Lambda T^{k_1} \Lambda \cdots \Lambda T^{k_n} \Lambda$, which vanishes by Theorem 1.1, and a word containing a vanishing subword vanishes. Hence $J^{n+1} = 0$. Conversely $(\Lambda T)^{n-1} \Lambda$ is a product of n elements of J and is the sharp word of Proposition 5.1, so $J^n \neq 0$.

Because J is a nilpotent two-sided ideal of a finite-dimensional algebra, $J \subseteq \text{rad } \mathcal{A}$ [23]. The quotient $\mathcal{A}/J$ is generated by the image of T , which satisfies $T^N = I$, so it is a quotient of $\mathbb{C}[x]/(x^N - 1)$. That algebra is semisimple, since $x^N - 1$ has N distinct roots over $\mathbb{C}$, and a quotient of a semisimple algebra is semisimple. Hence $\text{rad}(\mathcal{A}/J) = 0$, which combined with $J \subseteq \text{rad } \mathcal{A}$ gives $\text{rad } \mathcal{A}/J = 0$ and therefore $\text{rad } \mathcal{A} = J$.

For strictness, $I \notin J$ because J is nilpotent, so $\mathcal{A} \supsetneq J$. If $J^r = J^{r+1}$ for some $1 \leq r \leq n$ then iterating gives $J^r = J^{n+1} = 0$, contradicting $0 \neq J^n \subseteq J^r$. Finally, under the standard convention $\text{LL}(\mathcal{A}) = \min\{L \geq 1 : (\text{rad } \mathcal{A})^L = 0\}$, the two displayed facts give $\text{LL}(\mathcal{A}) = n + 1$. $\square$

The Loewy layers are $\mathcal{A}/J$, J/J^2 , $\dots$, J^{n-1}/J^n and J^n , and there are $n + 1$ of them, all nonzero. The number $n = N/2$ assigned to this algebra together with the conjecture [15] is the largest exponent r for which $J^r \neq 0$; it is one less than the number of layers, and Theorem 1.2 is what decides between the two readings, because it establishes both $J^n \neq 0$ and $J^{n+1} = 0$ rather than only one of them.

Remark 6.1. The decomposition of $\mathcal{A}$ into the span of the powers of T and the ideal generated by Λ , proposed with the conjecture, is confirmed by Theorem 1.2: the ideal is precisely the radical, and the quotient by it is the semisimple algebra generated by the image of T .

7. A FIVE-AXIOM FAMILY

Nothing in Sections 3 to 6 used the root of unity, the field $\mathbb{C}$, or the interpolation formulas themselves. Five properties carried the argument, and this section isolates them, proves the same theorems from them over an arbitrary field, and shows that none may be dropped.

Definition 7.1 (Chebyshev-filtered kaleidoscope datum). Let $\mathbb{k}$ be a field, E a finite-dimensional $\mathbb{k}$ -vector space, $n \geq 2$ an integer, and $R, \sigma \in \text{GL}(E)$. Put

$$C = R + R^{-1}, \quad D = \sigma(R - R^{-1}),$$

and, on $E \oplus E$,

$$\Theta = \begin{pmatrix} R^{-1} & 0 \\ R - R^{-1} & R \end{pmatrix}, \quad \Xi = \begin{pmatrix} 0 & \sigma \\ 0 & 0 \end{pmatrix}, \quad \mathcal{K} = \mathbb{k}\langle I, \Theta, \Xi \rangle, \quad \mathcal{J} = \mathcal{K} \Xi \mathcal{K}.$$

A *length- n Chebyshev-filtered kaleidoscope datum* is such a tuple together with a chain of subspaces $F_0, \dots, F_n$ of E obeying

- (A1) *exhaustive flag of width two*: $0 = F_0 \subset F_1 \subset \dots \subset F_n = E$ and $\dim F_r = 2r$;
- (A2) *propagation*: $CF_r \subseteq F_r$ for $0 \leq r \leq n$;
- (A3) *lowering*: $DF_r \subseteq F_{r-1}$ for $1 \leq r \leq n$;
- (A4) *survival*: $D^{n-1} \neq 0$;
- (A5) *semisimple cyclic carrier*: $B = \mathbb{k}[\Theta, \Theta^{-1}]$ is semisimple.

The operator C is the abstract counterpart of $X = t + t^{-1}$ and D of $A = s(t - t^{-1})$; the block matrices Θ and Ξ are what the similarity of Lemma 2.2 turns T and Λ into. Invertibility of R and σ is part of the data rather than a sixth axiom: it is what gives negative powers a meaning and what transports nonvanishing through the rightmost factor.

Over a field on which equality, linear algebra and polynomial factorization can be carried out, the five conditions are finite checks: (A1) from the ranks of the flag bases, (A2) and (A3) by reducing C and D applied to a flag basis against the target basis, (A4) from rank D^{n-1} , and (A5) — since Θ is invertible and hence $B = \mathbb{k}[\Theta]$ — from the absence of a repeated irreducible factor in the minimal polynomial of Θ . Over an arbitrary field they remain algebraic conditions with no claim of decidability.

Proposition 7.2 (The original model is a datum). *Let $N = 2n \geq 4$ be even and $u \in \mathcal{U}_N$. Take $\mathbb{k} = \mathbb{C}$, $E = V = \mathbb{C}^N$, $R = t$, $\sigma = s$ and $F_r = K_r$. Then all five axioms hold, $C = X$ and $D = A$, and $\Theta = S^{-1}TS$, $\Xi = S^{-1}\Lambda S$, so $\mathcal{K}$ is similar to $\mathcal{A}$ and $\mathcal{J}$ to J .*

Proof. The identifications of C , D , Θ and Ξ are Eq. (3.1) and Eq. (2.4). Axiom (A1) is Lemma 3.3, (A2) and (A3) are Lemma 3.4, and (A4) is Eq. (3.6) at $r = n - 1$, which gives $\text{rank } A^{n-1} = 2$. For (A5), $t^N = I$ gives $\Theta^N = I$ by Eq. (2.4), so the minimal polynomial of Θ divides $x^N - 1$, which is square-free over $\mathbb{C}$. $\square$

The embedding is uniform: it holds at every even N and every admissible u at once, with no genericity condition.

Lemma 7.3 (Block powers and the sandwich). *With q_k as in Eq. (4.1), for every $k \in \mathbb{Z}$*

$$R^k - R^{-k} = (R - R^{-1}) q_k(C), \quad \Theta^k = \begin{pmatrix} R^{-k} & 0 \\ R^k - R^{-k} & R^k \end{pmatrix},$$

and consequently

$$\Xi \Theta^k \Xi = \begin{pmatrix} 0 & D q_k(C) \sigma \\ 0 & 0 \end{pmatrix}. \quad (7.1)$$

Proof. The first identity is Lemma 4.1 with t replaced by R ; its proof used only that R commutes with R^{-1} , and the recurrence Eq. (4.1) has integer coefficients, so it is valid over any field. The second holds for $k = 0, 1$ and propagates: multiplying the displayed matrix by Θ sends the lower-left block to $(R^k - R^{-k})R^{-1} + R^k(R - R^{-1}) = R^{k+1} - R^{-(k+1)}$ and the diagonal blocks to $R^{-(k+1)}$ and R^{k+1} . Block inversion gives $\Theta^{-1} = \begin{pmatrix} R & 0 \\ R^{-1} - R & R^{-1} \end{pmatrix}$, and the same induction covers $k < 0$. Multiplying out $\Xi \Theta^k \Xi$ leaves only the upper-right block, which is $\sigma(R^k - R^{-k})\sigma = D q_k(C)\sigma$. $\square$

Equation (7.1) fixes the order of the factors: the lowering operator D stands to the left of the polynomial $q_k(C)$, not to its right.

Theorem 7.4 (Axiomatic threshold and sharpness). *Let $(E, R, \sigma, F_\bullet)$ be a length- n Chebyshev-filtered kaleidoscope datum. Then*

$$\Xi \Theta^{k_1} \Xi \dots \Xi \Theta^{k_M} \Xi = 0 \quad \text{for all } M \geq n, (k_1, \dots, k_M) \in \mathbb{Z}^M, \quad (7.2)$$

while at $M = n - 1$ and $k_1 = \dots = k_{n-1} = 1$ the same word is nonzero. Only (A1), (A2) and (A3) are used for the vanishing, and (A4) only for the nonvanishing.

Proof. Iterating Eq. (7.1), the only possibly nonzero block of the word in Eq. (7.2) is the upper-right one, $Dq_{k_1}(C)Dq_{k_2}(C)\cdots Dq_{k_M}(C)\sigma$. By (A2) every polynomial in C preserves every F_r , so by (A3) each composite $Dq_k(C)$ carries F_r into F_{r-1} . Since σ is invertible, $\sigma E = E = F_n$ by (A1). Reading the product from the right, $M \geq n$ lowering steps land in $F_0 = 0$. For the sharp word $q_1 = 1$, so the upper-right block is $D^{n-1}\sigma$, nonzero by (A4) and the surjectivity of σ . $\square$

Theorem 7.5 (Radical and layers from the axioms). *For every length- n Chebyshev-filtered kaleidoscope datum,*

$$\mathcal{J} = \text{rad } \mathcal{K}, \quad \mathcal{J}^{n+1} = 0, \quad \mathcal{J}^n \neq 0,$$

every inclusion in $\mathcal{K} \supsetneq \mathcal{J} \supsetneq \cdots \supsetneq \mathcal{J}^n \supsetneq \mathcal{J}^{n+1} = 0$ is strict, and $\mathcal{K}$ has exactly $n+1$ nonzero Loewy layers.

Proof. The argument of Theorem 1.2 applies verbatim with Theorem 7.4 in place of Theorem 1.1: a product of $n+1$ elements of $\mathcal{J}$ expands into words with $n+1$ copies of Ξ separated by powers of Θ , each containing a vanishing subword; and $(\Xi\Theta)^{n-1}\Xi$ is a nonzero product of n elements of $\mathcal{J}$. For the radical, $\mathcal{K}/\mathcal{J}$ is generated by the image of Θ , so the natural map $B \rightarrow \mathcal{K}/\mathcal{J}$ is surjective; axiom (A5) makes B semisimple, hence so is its quotient, and the two inclusions close as before. Strictness follows from $\mathcal{J}^n \neq 0$ exactly as in Theorem 1.2. $\square$

No formula for the dimension of an individual layer $\mathcal{J}^r/\mathcal{J}^{r+1}$ is claimed; the axioms fix how many layers there are and that each is nonzero, not how large each one is.

Every axiom is needed. Each of the five axioms is now shown to be independent: for each one there is a finite-dimensional datum over $\mathbb{Q}$ that satisfies the other four and violates a conclusion the deleted axiom supports. All five constructions are exact, with no fitted parameter and no genericity assumption.

Two conventions keep the examples small. When base data R_0, σ_0 on E_0 and a base flag $G_0 \subset \cdots \subset G_n$ are displayed, the datum is the doubling $E = E_0 \oplus E_0$, $R = R_0 \oplus R_0$, $\sigma = \sigma_0 \oplus \sigma_0$ and $F_r = G_r \oplus G_r$, which turns $\dim G_r = r$ into the width $\dim F_r = 2r$ that (A1) demands. And for any invertible R the matrix S of Eq. (2.3) conjugates $\text{diag}(R, R^{-1})$ into Θ , so (A5) holds whenever R is diagonalizable with a square-free minimal polynomial; in particular it holds for every diagonal R_0 with distinct nonzero entries.

Theorem 7.6 (Independence). *For each axiom in $\{(A1), \dots, (A5)\}$ there is a finite-dimensional datum over $\mathbb{Q}$ satisfying the other four and falsifying a conclusion of Theorem 7.4 or Theorem 7.5 that the deleted axiom supports.*

Proof. We give one witness for each axiom.

Deleting (A1). Let $n = 2$, $E_0 = \mathbb{Q}^3$,

$$R_0 = \text{diag}(1, 2, 3), \quad \sigma_0 = \begin{pmatrix} 0 & 2/3 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 3/8 \end{pmatrix},$$

and $G_0 = 0$, $G_1 = \langle e_1 \rangle$ and $G_2 = \langle e_1, e_2 \rangle$, a proper subspace of E_0 . Then $C_0 = \text{diag}(2, 5/2, 10/3)$ and $D_0 = \sigma_0 \text{diag}(0, 3/2, 8/3)$ has $D_0 e_1 = 0$, $D_0 e_2 = e_1$ and $D_0 e_3 = e_3$. Axiom (A2) holds because C_0 is diagonal, (A3) because $D_0 G_1 = 0$ and $D_0 G_2 \subseteq G_1$, (A4) because $D \neq 0$, and (A5) because R_0 is diagonal with distinct entries. Only exhaustivity fails, and $D_0^2 e_3 = e_3$, so $D^2 \sigma \neq 0$ and the threshold word at $M = n = 2$ survives.

Deleting (A2). Let $n = 3$, $E_0 = \mathbb{Q}^3$, $v = e_2 + e_3$,

$$R_0 = \text{diag}(1, 2, 3), \quad \sigma_0 = \begin{pmatrix} 0 & 2/3 & 0 \\ 1 & -2/3 & 3/8 \\ -1 & -2/3 & 3/8 \end{pmatrix},$$

and $G_1 = \langle e_1 \rangle$, $G_2 = \langle e_1, v \rangle$, $G_3 = E_0$. Then $C_0 = \text{diag}(2, a, b)$ with $a = 5/2$ and $b = 10/3$, and $D_0 e_1 = 0$, $D_0 e_2 = e_1 - v$, $D_0 e_3 = v$. Axioms (A1), (A3), (A4) and (A5) hold; in particular $D_0 v = e_1$ and $D_0^2 e_3 = e_1 \neq 0$. But $C_0 v = a e_2 + b e_3 \notin G_2$ because $a \neq b$, so (A2) fails. With $\delta = b - a = 5/6$ and $q_2(x) = x$, a direct computation gives

$$(D_0 C_0)^3 e_3 = b \delta (a e_1 + \delta v) \neq 0,$$

so the threshold tuple $(2, 2, 2)$ survives at $M = n = 3$. The failure is exactly the one (A2) forbids: the polynomial $q_2(C)$ pushes a vector back up the flag faster than D pulls it down.

Deleting (A3). Let $n = 2$, $R_0 = \text{diag}(2, 3)$, $\sigma_0 = I_2$, $G_1 = \langle e_1 \rangle$ and $G_2 = E_0$. Then $C_0 = \text{diag}(5/2, 10/3)$ and $D_0 = \text{diag}(3/2, 8/3)$, so (A1), (A2), (A4) and (A5) hold while $D_0 G_1 = G_1 \not\subseteq G_0 = 0$. Since $D_0^2 \neq 0$, the threshold word at $M = 2$ survives.

Deleting (A4). Let $n = 3$, $E = \mathbb{Q}^6$, $R = \sigma = I_6$ and $F_r = \langle e_1, \dots, e_{2r} \rangle$. Then $C = 2I$, $D = 0$ and $\Theta = I$, so (A1), (A2), (A3) and (A5) hold while $D^{n-1} = 0$. Here $\mathcal{K} = \mathbb{Q}I \oplus \mathbb{Q}\Xi$ and $\mathcal{J} = \mathbb{Q}\Xi$ with $\mathcal{J}^2 = 0$, so the Loewy length is 2 rather than $n + 1 = 4$: without (A4) the layer count of [Theorem 7.5](#) collapses.

Deleting (A5). Let $n = 2$, put $\mathcal{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$,

$$R_0 = I_2 + \mathcal{N}, \quad \sigma_0 = \frac{1}{2}I_2,$$

and $G_1 = \ker \mathcal{N} = \langle e_1 \rangle$, $G_2 = E_0$. Since $\mathcal{N}^2 = 0$ one has $R_0^{-1} = I_2 - \mathcal{N}$, hence $C_0 = 2I$ and $D_0 = \mathcal{N}$, so (A1)–(A4) hold. The minimal polynomial of Θ is $(x - 1)^2$, so (A5) fails. Write $H = \Theta - I$; on one summand of the doubling

$$H = \begin{pmatrix} -\mathcal{N} & 0 \\ 2\mathcal{N} & \mathcal{N} \end{pmatrix}, \quad \Xi = \begin{pmatrix} 0 & \frac{1}{2}I_2 \\ 0 & 0 \end{pmatrix},$$

and $H^2 = \Xi^2 = H\Xi H = 0$. Those three relations close the algebra:

$$\mathcal{K} = \text{span}_{\mathbb{Q}}\{I, H, \Xi, H\Xi, \Xi H, \Xi H\Xi\}, \quad \mathcal{J} = \text{span}_{\mathbb{Q}}\{\Xi, H\Xi, \Xi H, \Xi H\Xi\}.$$

Every spanning matrix of $\mathcal{J}$ has zero lower-left block, whereas the lower-left block of H is $2\mathcal{N} \neq 0$; hence $H \notin \mathcal{J}$. The class of H in $\mathcal{K}/\mathcal{J}$ is then nonzero with square zero, so $\text{rad}(\mathcal{K}/\mathcal{J}) \neq 0$ and $\mathcal{J} \subsetneq \text{rad} \mathcal{K}$: without (A5) the ideal generated by Ξ is still nilpotent but is no longer the whole radical. The doubled datum is two faithful copies of this algebra, so the same ideal-membership argument applies to it. $\square$

Each witness removes one axiom and keeps the other four, so [Theorem 7.6](#) is a statement about necessity and not merely a list of ways the hypotheses can fail. Read together with [Proposition 7.2](#), it says that the five conditions are a minimal description of what makes the kaleidoscope algebra behave as [Theorems 1.1](#) and [1.2](#) say it does.

8. INDEPENDENT EXACT CHECKS

The results above are proved without computation, and the reader can follow every step from the page. Exact computations were nevertheless carried out against the definitions, as a check on the conventions — the direction of the shift, the order of the factors in a word, the treatment of negative and zero exponents, and the behaviour at parameters where nodes of opposite parity collide. They are reported here because each one is a way the argument could have been wrong without being visibly wrong.

Reconstructing t , d , s , T and Λ over the field with 97 elements, which contains primitive 4th, 6th and 8th roots of unity, one can enumerate threshold words exhaustively rather than sample them. For $N = 4$ and $N = 6$ every admissible spectral parameter in that field was used — 92 and 90 values respectively, the excluded ones being the roots of $u^N = 1$ — together with every exponent tuple $(k_1, \dots, k_n)$ with $0 \leq k_i < N$, giving 16 and 216 tuples per parameter. For $N = 8$ all 4096 tuples were used at three admissible parameters, one of them satisfying $u^8 = -1$, where the even

and odd node sets intersect while each remains internally distinct, as [Lemma 3.1](#) allows. The three families comprise 33,200 words. Every one vanished, and at every parameter the all-ones word one step below the threshold was nonzero of rank two.

A separate computation in characteristic zero, over exact algebraic extensions, constructed the sharp word $A^{n-1}s$ for $N = 4, 6, 8$. In each case its rank is two, the radical of the common denominator of its entries is $u^N - 1$, and its nonzero entries have no common zero in $\mathbb{C}^\times$; these are [Propositions 5.1](#) and [5.2](#) at three sizes, obtained from the matrix definitions rather than from the interpolation argument.

For the axiomatic family, the block-power and sandwich identities of [Lemma 7.3](#) were checked to have vanishing residual for every exponent from -3 to 3 and for mixed tuples of positive, zero and negative exponents, with a non-diagonal R and a σ that does not commute with it. In one explicit datum over $\mathbb{Q}$ of length $n = 2$, all 25 threshold words with exponents in $\{-2, \dots, 2\}$ vanish while the sharp word has rank two. The five witnesses of [Theorem 7.6](#) were rebuilt from the displayed matrices and each was found to satisfy exactly four of the five axioms; the three surviving threshold words have ranks 2, 2 and 4, the datum deleting (A4) has Loewy length 2, and the datum deleting (A5) has $\dim \mathcal{K} = 6$, $\dim \mathcal{J} = 4$, $\dim \mathcal{J}^2 = 1$, $\mathcal{J}^3 = 0$ and $H \notin \mathcal{J}$.

None of these computations is a premise of any statement above. They rule out the transcription and convention errors that a purely symbolic argument cannot rule out about itself, and they agree with it everywhere.

9. CONCLUSION

Words in the kaleidoscope Yang–Baxter algebra die at a threshold set by half the clock order, and they do so for one reason: the operator $s(t - t^{-1})$ is a divided difference in the variable $y_j = u\omega^j + u^{-1}\omega^{-j}$ and therefore lowers polynomial degree, $t + t^{-1}$ is a symmetric function of the same two shifts and therefore does not raise it, and the integer Chebyshev identity $t^k - t^{-k} = (t - t^{-1})q_k(t + t^{-1})$ makes every intervening power of T act like the first one. Since the two parity classes carry $n = N/2$ distinct nodes each, n lowering steps exhaust the flag and every word with n intervening powers vanishes, for every even N , every admissible spectral parameter and every tuple of integer exponents. One step fewer leaves a matrix of rank two whose poles are exactly the N th roots of unity and which vanishes at no admissible parameter, so the threshold is sharp in the strongest available sense. The ideal generated by Λ is therefore the Jacobson radical, its $(n+1)$ st power vanishes and its n th does not, and the algebra has $n+1 = N/2+1$ Loewy layers — one more than the value that was proposed with the conjecture, which counts instead the largest nonvanishing radical power.

The mechanism is not tied to the root of unity or to the complex field. Five conditions — an exhaustive flag of width two, a propagating symmetric operator, a lowering operator, survival of its $(n-1)$ st power, and a semisimple cyclic carrier — reproduce the threshold, the sharpness and the radical over an arbitrary field, and each of the five is necessary, as an explicit rational counterexample for the deletion of each one shows.

What lies immediately beyond is the size rather than the number of the layers: the axioms fix that there are $n+1$ nonzero Loewy quotients but say nothing about the dimension of each, and a classification of the data satisfying them is open. Odd clock order, spectral parameters on the excluded circle, other representations of the kaleidoscope Yang–Baxter relation, and the consequences of the filtration for the spectrum of the underlying scattering problem all lie outside the argument given here. A single admissible exponent tuple at the threshold with a nonzero word, or a single admissible parameter at which the all-ones word one step below it vanishes, would refute what is proved here; none exists.

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